

1. Violet light is incident on a target with a work function of 3.68×10^{-19} joules. The maximum kinetic energy of an electron ejected from the target is given by $K_{\max} = \frac{hc}{\lambda} - \phi$. $K_{\max}$ is the maximum kinetic energy of the ejected electron in joules, h is Planck's constant which equals 6.626×10^{-34} joule-seconds, c is the speed of light which equals 3×10^8 meters per second, λ is the wavelength of the incident light in meters, and ϕ is the work function of the target in joules. If the maximum kinetic energy of an ejected electron equals 1.05×10^{-19} joules, the wavelength of the incident light equals _____ $\times 10^{-9}$ m.
- (A) 390 (B) 400 (C) 410 (D) 420 (E) 430
2. Consider the line that contains points $A(-12, -6)$ and $B(8, 10)$. If the x-intercept of $\overline{AB}$ is the point $E(a, 0)$ and the y-intercept of $\overline{AB}$ is the point $F(0, b)$, then $a + b =$ _____. (nearest tenth)
- (A) -1.5 (B) -1.2 (C) -0.9 (D) -0.6 (E) -0.3
3. Rose's Cookie Shoppe is open from 2:00 to 6:00. She sells sugar cookies for \$0.65 each and chocolate chip cookies for \$1.15 each. On Thursday, Rose sold a total of 174 cookies and netted \$140.10. How many sugar cookies did she sell?
- (A) 112 (B) 114 (C) 116 (D) 118 (E) 120
4. The Grand Oaks High School Math Booster Club flew in Sounds Like Reign to perform at a fundraiser. Each ticket for a seat in rows 1 through 5 cost \$65. Each ticket for a seat in rows 6 through 20 cost \$50. Each ticket for a seat in rows 21 through 50 cost \$35. A total of 1,100 tickets were sold and a total of \$46,750 was netted. They sold six times as many tickets for rows 21 through 50 as they sold for rows 1 through 5. How many tickets did they sell for rows 6 through 20?
- (A) 326 (B) 330 (C) 334 (D) 338 (E) 342
5. Stephen and Jonathan work for the Dallas ISD each summer. Stephen can paint a classroom by himself in 12 hours. Jonathan can paint a classroom by himself in 15 hours. If they work together, they can paint 48 classrooms in _____ hours.
- (A) 320 (B) 322 (C) 324 (D) 326 (E) 328
6. Heberto is planning to take six number sense tests this week. If he scores 327, 341, 313, 335 and 319 on the first five tests, what score does he need to make on the sixth test in order to have an overall average of 333?
- (A) 451 (B) 354 (C) 357 (D) 360 (E) 363
7. Rancher Rob has a water trough shaped like a trapezoidal prism. Each trapezoidal end of the trough has a height of 3 ft 6 in, a top base of 4 ft 4 in, and a lower base of 3 ft 9 in. The length of the trough is 6 ft 6 in. How many gallons of water does the trough hold when it is completely full? (nearest gallon)
- (A) 688 (B) 692 (C) 696 (D) 700 (E) 704

8. During the summer, Dennis traveled from his home in Salado to his favorite vacation campground in Grand Teton National Park. On day 1, he traveled 612 miles at an average speed of 64 mph. On day 2, he traveled at an average speed of 68 mph. On day 3, he traveled 428 miles at an average speed of 72 mph. If the total drive time was 23 hours 11 minutes, how far did he travel on day 2? (nearest whole number)

- (A) 514 mi (B) 518 mi (C) 522 mi (D) 526 mi (E) 530 mi

9. The area of Patrick's square-shaped lot outside Whitehouse is 5 acres. How long is each outer edge of the lot? (nearest foot)

- (A) 458 ft (B) 461 ft (C) 464 ft (D) 467 ft (E) 470 ft

10. Richmond Randle played Dallas South Oak Cliff in the 5A division 2 state championship game in 2024. The two teams combined for 73 points and Randle won by a margin of 3 points. How many points did the Oak Cliff score?

- (A) 34 (B) 35 (C) 36 (D) 37 (E) 38

11. Find the area of quadrilateral ABCD.

- (A) 124 (B) 126 (C) 128
(D) 130 (E) 132

12. Draw auxiliary line segment $\overline{BD}$. Find the perimeter of $\triangle BDC$. (nearest tenth)

- (A) 41.6 (B) 41.9 (C) 42.2
(D) 42.5 (E) 42.8

13. If P be the midpoint of $\overline{AB}$ and Q be the midpoint of $\overline{CD}$, then $PQ =$ _____. (nearest tenth)

- (A) 12.0 (B) 12.2 (C) 12.4 (D) 12.6 (E) 12.8

14. $m\angle ADC =$ _____. (nearest whole number)

- (A) 120 (B) 122 (C) 124 (D) 126 (E) 128

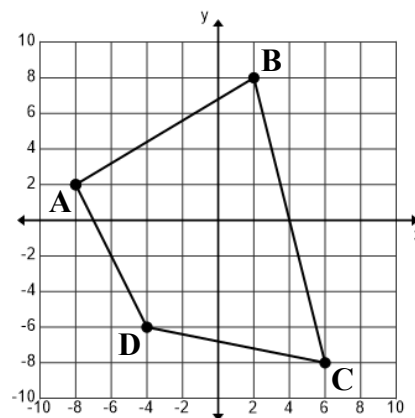
15-16. The vertices of $\triangle EFG$ are $E(-5, 3)$, $F(1, 9)$ and $G(13, -5)$. Point $H(e, f)$ is the centroid of the triangle and point J is the midpoint of $\overline{EG}$ and point K is the midpoint of $\overline{FG}$.

15. $f + e =$ _____. (nearest tenth)

- (A) 4.9 (B) 5.1 (C) 5.3 (D) 5.5 (E) 5.7

16. The area of $\triangle JKG =$ _____. (nearest tenth)

- (A) 19.5 (B) 19.8 (C) 20.1 (D) 20.4 (E) 20.7

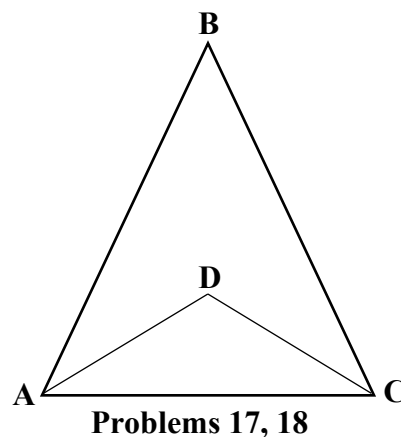


Problems 11, 12, 13, 14

17-18. Consider isosceles triangle ABC with $AB = 32$ and $CB = 32$.

$\overline{AD}$ bisects $\angle BAC$ and $\overline{CD}$ bisects $\angle BCA$.

$m\angle ABC = 36^\circ$.



17. $m\angle ADC =$ _____. (nearest whole number)

- (A) 106°
- (B) 108°
- (C) 110°
- (D) 112°
- (E) 114°

18. Find the area of quadrilateral $BADC$.
(nearest whole number)

- (A) 221
- (B) 224
- (C) 227
- (D) 230
- (E) 233

19. If $f(x) = \frac{-5+3x}{6-4x}$, what is the value of c if $f^{-1}(c) = 1$?

- (A) -5
- (B) -4
- (C) -3
- (D) -2
- (E) -1

20. The 7th term of an arithmetic sequence is 45 and the 11th term is 73. Find the sum of the first 20 terms.

- (A) 1380
- (B) 1390
- (C) 1400
- (D) 1410
- (E) 1420

21-22. Thomas has a large jar full of 545 marbles. All of these are either red, blue or green. There are thirty more reds than blues and there are 11 more green marbles than blue marbles.

21. If Thomas randomly selects 3 marbles without replacement, what is the probability that all 3 will be blue? (nearest ten-thousandth)

- (A) 0.0277
- (B) 0.0281
- (C) 0.0285
- (D) 0.0289
- (E) 0.0293

22. If Thomas randomly selects 4 marbles with replacement, what is the probability that none of the 4 will be red? (nearest thousandth)

- (A) 0.160
- (B) 0.164
- (C) 0.168
- (D) 0.172
- (E) 0.176

23. If $f(x) = \sqrt{3x+10}$, $g(x) = x^2 + 4$, $(f \circ g)(c) = 7$, and $c > 0$, then $c =$ _____.

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

24. A population of Amazonian Army Ants triples every 64 days. At $t = 0$, there were 24 ants. The population is expected to reach 5832 at $t =$ _____ days. (nearest whole number)

- (A) 314
- (B) 316
- (C) 318
- (D) 320
- (E) 322

25-26. Consider the sequence $1, \frac{5}{6}, \frac{5}{12}, \frac{17}{120}, \frac{13}{360}, \frac{37}{5040}, \dots$

25. The next term in the sequence is _____. (nearest hundred-thousandth)

- (A) 0.00118 (B) 0.00121 (C) 0.00124 (D) 0.00127 (E) 0.00130

26. The sum of the first 20 terms of the sequence is _____. (nearest hundred-thousandth)

- (A) 2.43653 (B) 2.43656 (C) 2.43659 (D) 2.43662 (E) 2.43665

27. Assume that the average low temperature in Idaho Falls in February is 16°F at 5:00 AM, and the average high temperature is 32°F at 5:00 PM. Assume the temperature varies sinusoidally each day. On a typical day in February, at what time does the morning temperature reach 25°F ? (nearest minute)

- (A) 11:13 AM (B) 11:17 AM (C) 11:21 AM (D) 11:25 AM (E) 11:29 AM

28. How many odd three digit counting numbers less than 600 are there such that the second digit cannot be 2, 4 or 6?

- (A) 166 (B) 169 (C) 172 (D) 175 (E) 178

29-30. Consider an ellipse with vertices $(2, 6)$ and $(10, 6)$, and with a minor axis of length 4.

29. Find the eccentricity of the ellipse. (nearest hundredth)

- (A) 0.75 (B) 0.78 (C) 0.81 (D) 0.84 (E) 0.87

30. Find the area of the ellipse. (nearest tenth)

- (A) 25.1 (B) 25.4 (C) 25.7 (D) 26.0 (E) 26.3

31. The measure of an interior angle of a regular octagon is _____ $^{\circ}$.

- (A) 105 (B) 120 (C) 135 (D) 150 (E) 165

32-33. Consider $\triangle ABC$ vertices $A(2, -3, 5)$, $B(-3, 2, -1)$ and $C(-5, -4, 7)$.

32. Find the perimeter of $\triangle ABC$. (nearest tenth)

- (A) 26.2 (B) 26.4 (C) 26.6 (D) 26.8 (E) 27.0

33. The length of the shortest median in $\triangle ABC$ is _____. (nearest tenth)

- (A) 6.2 (B) 6.4 (C) 6.6 (D) 6.8 (E) 7.0

34. Consider a set of consecutive positive even integers beginning with 6 and ending with n . If one of the integers is removed, the average of the remaining integers is $46\frac{22}{41}$. Which integer is removed?

- (A) 60 (B) 62 (C) 64 (D) 66 (E) 68

35-36. Regular hexagon ABCDE has an area of 842.

35. The area of the circle inscribed in the hexagon = _____. (nearest whole number)

- (A) 755 (B) 758 (C) 761 (D) 764 (E) 767

36. The area of the circle circumscribed about the hexagon = _____. (nearest whole number)

- (A) 1002 (B) 1006 (C) 1010 (D) 1014 (E) 1018

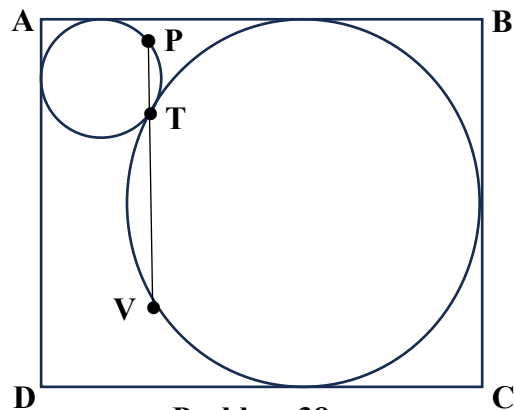
37. If $\cos \theta = \frac{12}{13}$ and θ is in quadrant IV, and if $\tan \alpha = -\frac{7}{24}$ and α is in quadrant II, then what is the value of $\sin(\theta + \alpha)$?

- (A) $\frac{8}{13}$ (B) $\frac{202}{325}$ (C) $\frac{204}{325}$ (D) $\frac{206}{325}$ (E) $\frac{16}{25}$

38. Given: rectangle ABCD, $AD = 52.8$, $PV = 39.6$.

$\overline{PV}$ is parallel to $\overline{BC}$. The circles are tangent to the sides of the rectangle and to each other. Point T lies on $\overline{PV}$ and is a point of tangency. Draw auxiliary line segment $\overline{DB}$. $DB =$ _____. (nearest tenth)

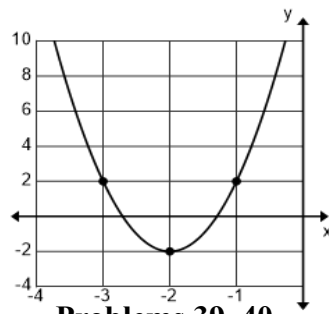
- (A) 78.6 (B) 78.9 (C) 79.2
(D) 79.5 (E) 79.8



Problem 38

39. The equation of the directrix of the parabola is $y =$ _____. (nearest tenth)

- (A) $-\frac{33}{16}$ (B) -4 (C) -3
(D) $-\frac{9}{4}$ (E) $-\frac{17}{8}$



Problems 39, 40

40. The length of the latus rectum of the parabola is _____.

- (A) 0.125 (B) 0.25 (C) 0.5
(D) 1 (E) 2

41-43. Given: $h(x) = 6e^{(0.2x)}$ and $f(x) = 60$.

41. If the x-intercept of the line tangent to the graph of $y = h(x)$ when $x = 10$ is the point $P(a, 0)$, then $a =$ _____. (nearest tenth)

- (A) 4.8 (B) 5.0 (C) 5.2 (D) 5.4 (E) 5.6

42. Consider the first quadrant region bounded by the y-axis, the graphs of $y = f(x)$ and the graph $y = h(x)$. Find the area of this region. (nearest whole number)

- (A) 409 (B) 412 (C) 415 (D) 418 (E) 421

43. Consider the first quadrant region bounded by the y-axis, the graph of $y = f(x)$ and the graph $y = h(x)$. If this region is revolved around the line $x = 15$, the volume of the solid of revolution equals _____. (nearest whole number)

- (A) 27,892 (B) 27,896 (C) 27,900 (D) 27,904 (E) 27,908

44. There are two values in the interval $[2\pi, 4\pi]$ that satisfy the conclusion of the Mean Value Theorem for $f(x) = 4\sin\left(\frac{x}{6} - \frac{\pi}{2}\right) - \frac{x}{4}$. The sum of these values is _____. (nearest hundredth)

- (A) 18.52 (B) 18.63 (C) 18.74 (D) 18.85 (E) 18.96

45. A person is standing 400 feet away from a model rocket on the ground. The rocket is fired straight up at a constant speed of 25 ft/s. Find the rate at which the distance between the person and the rocket is increasing 30 seconds after liftoff. (nearest whole number)

- (A) 18 ft/s (B) 19 ft/s (C) 20 ft/s (D) 21 ft/s (E) 22 ft/s

46. An engineer is designing a box shaped like a rectangular prism with a square base. The box is required to have a volume of 500 cm^3 . Find the minimum total surface area of a box that fits the specified criteria. (The box has a top.) (nearest whole number)

- (A) 372 cm^2 (B) 375 cm^2 (C) 378 cm^2 (D) 481 cm^2 (E) 384 cm^2

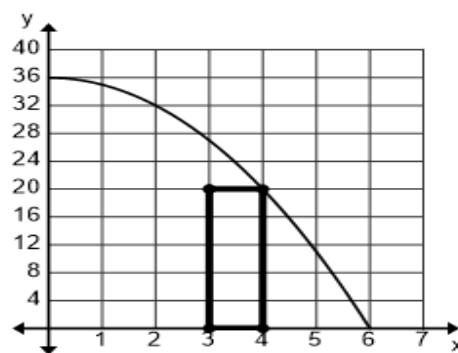
47. Given: $3x^2 + 2xy^2 = 51$. Find the value of $\frac{dy}{dx}$ at the point $(3, b)$, $b < 0$. (nearest hundredth)

- (A) 1.05 (B) 1.08 (C) 1.11 (D) 1.14 (E) 1.17

48. A particle moving along the x-axis has velocity $v(t) = 3(t + e^{-3t})\cos(0.4t)$ for $0 \leq t \leq 12$. Find the maximum speed of the particle over $0 \leq t \leq 12$. (nearest tenth)

- (A) 8.0 (B) 22.1 (C) 36.2 (D) 50.3 (E) 64.4

49. Consider the graph shown on the right. Use the Right Rectangle Approximation Method with three subintervals of equal width to approximate the area bounded by the curves $y_1 = 36 - x^2$, $y_2 = 0$, $x_1 = 2$, $x_2 = 5$. One of the rectangles is shown on the right. (nearest tenth)



Problem 49

- (A) 58.0 (B) 58.3 (C) 58.6
(D) 58.9 (E) 60.2

50. Consider the graph of the polar equation $r = 3 \sin(2\theta)$. Find the area of one of the rose petals. (nearest tenth)

- (A) 3.2 (B) 3.5 (C) 3.8 (D) 4.1 (E) 4.4

- 51-54. Suppose that on June 1st, 2018, 100 endangered miniature frogs were released into an area in Mexico conducive to their survival. On June 1st, 2020, the population had reached 160. Dr. Smith, a researcher from UT Arlington, determined that the area could support no more 2,000 of the miniature frogs. He developed a logistic equation to model the rate of growth of the frog population.

51. The researcher's equation predicts that the rate of growth of the population of miniature frogs will reach a maximum of _____ frogs per year. (nearest tenth)

- (A) 123.3 (B) 124.4 (C) 125.5 (D) 126.6 (E) 127.7

52. The solution to the researcher's equation predicts the population will reach 672 in June of _____.

- (A) 2027 (B) 2028 (C) 2029 (D) 2030 (E) 2031

	Mass – mean	Mass – Stan Dev	Length – mean	Length – Stan Dev
Males	0.22 g	0.04 g	12.7 mm	0.84 mm
Females	0.16 g	0.03 g	10.8 mm	0.72 mm

- 53-54. The year the population reached 400, Dr. Smith spent the month of July recording the weights of 26 randomly selected adult males and 24 randomly selected adult females. A plot of the data revealed that the distribution of the masses of the male frogs was approximately normal and that the distribution of the masses of the female frogs was also approximately normal. His results are in the table above.

53. One of the adult males had a measured length of 13.8 mm. What percentile did that place him at?

- (A) 90th (B) 92nd (C) 94th (D) 96th (E) 98th

54. Consider the random variable D, the difference in the masses of the male and female miniature frogs. The mean of D is 0.06 g and the standard deviation of D is _____ g. (nearest hundredth)

- (A) 0.03 (B) 0.04 (C) 0.05 (D) 0.06 (E) 0.07

Runner	Karen	Melissa	Emma	Ronita	Cindy	Mary	Terri
Laps	32	22	30	34	36	41	37

55-56. In 1974, the Portland Track Club hosted a one-hour run on the 440-yd track at the Mount Hood Community College track. Seven ladies competed in the 20-29 age group. The number of laps each runner completed is in the table above.

55. A modified boxplot of the data revealed _____ outliers.

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

56. The difference between the median and mean is _____. (nearest hundredth)

- (A) 0.74 (B) 0.77 (C) 0.80 (D) 0.83 (E) 0.86

Average Weekly Mileage	36	44	60	50	56	46
10-K Time	33 min 36 sec	32 min 26 sec	30 min 32 sec	31 min 24 sec	30 min 44 sec	32 min 12 sec

57-58. During the late 70s, Joe entered six 10-K races. He compared his times with the average weekly mileage he put in the six weeks leading up to the race. He plotted the data and decided that there was a linear relationship between mileage and time. Then he calculated a LSRL for the data.

57. The LSRL showed that on average, for each additional mile he runs each week, there is a corresponding decrease of _____ seconds in his 10-K time. (nearest whole number)

- (A) 8 (B) 10 (C) 12 (D) 14 (E) 16

58. According to the LSRL, if Joe hoped to run a sub 29-minute 10-K, the minimal weekly mileage he would need to average is _____ miles per week. (nearest whole number)

- (A) 66 (B) 70 (C) 74 (D) 78 (E) 82

X	1	2	3	4	5	6
P(X)	.12	.13	.16	.17	.19	.23

59-60. Elise and Caleb are playing a game with an old weighted spinner. Let X be the value obtained when using the game's spinner. The probabilities are shown in the table above. Use the table for problems 59 and 60.

59. Find the expected value of a spin. (nearest hundredth)

- (A) 3.81 (B) 3.83 (C) 3.85 (D) 3.87 (E) 3.89

60. Find the standard deviation of a spin. (nearest hundredth)

- (A) 1.65 (B) 1.67 (C) 1.69 (D) 1.71 (E) 1.73